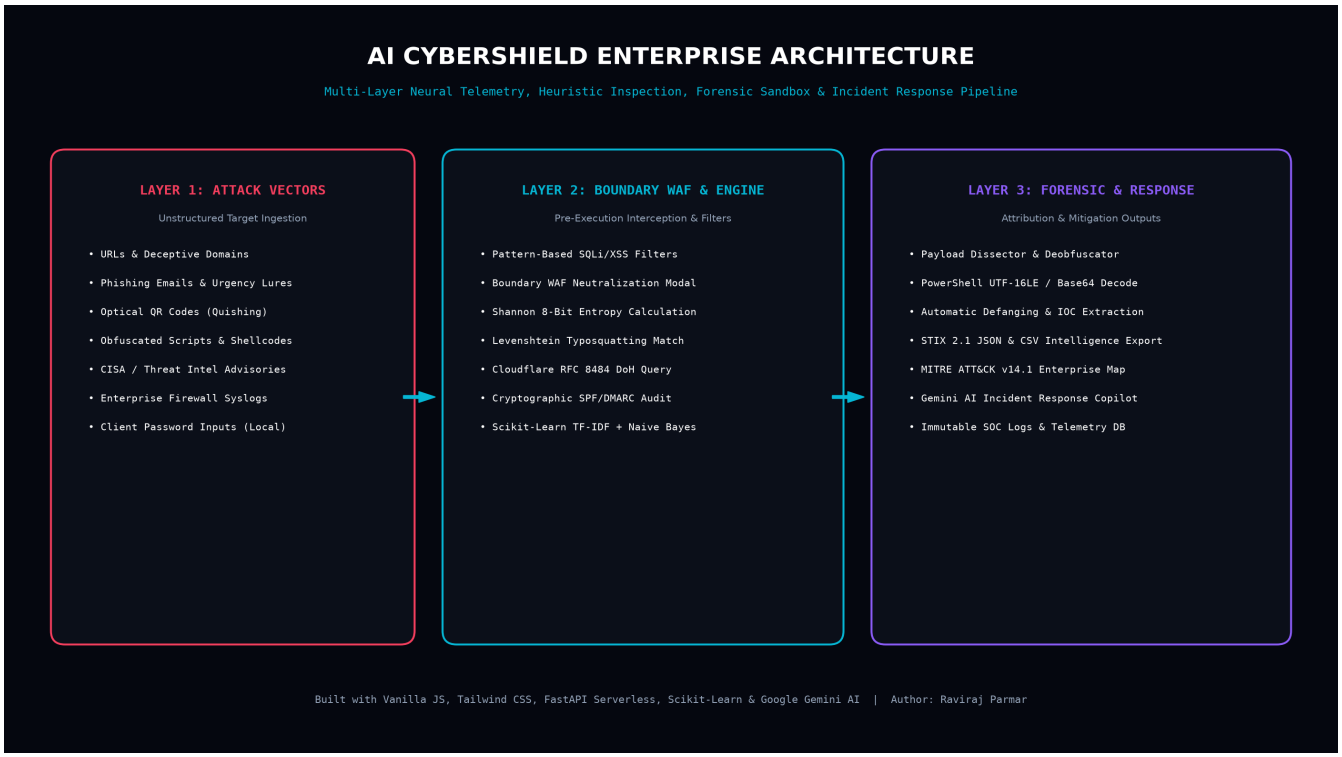
AI CYBERSHIELD

NEXT-GENERATION ENTERPRISE SOC & THREAT DEFENSE PLATFORM

A Comprehensive Technical Dossier, Architectural Blueprint & Oral Defense Guide
Live Production System: https://cybershield-cyber-shield-raviraj-parmar.vercel.app/

Project Title	AI CyberShield: Enterprise SOC & Cyber Defense Suite
Author / Lead	Raviraj Parmar
Core Capabilities	Multi-Vector Threat Scanning, RFC 8484 DoH Forensics, Payload Deobfuscator, Threat Intel STIX 2.1 IOC Extractor, MITRE ATT&CK; Matrix, Cryptographic Entropy Engine, Active WAF
Technology Stack	Python FastAPI, Vercel Serverless, Vanilla JS (ES6+), Scikit-Learn ML, Google Gemini AI, Tailwind CSS, Canvas API
Target Audience	Academic Evaluators, Technical Recruiters, SOC Analysts, Cybersecurity Students
Security Standard	Zero-Trust Architecture, RFC 8484 DoH, NIST NVD Integration, MITRE ATT&CK; v14.1 Enterprise Taxonomy


AI CYBERSHIELD ENTERPRISE ARCHITECTURE
Multi-Layer Neural Telemetry, Heuristic Inspection, Forensic Sandbox & Incident Response Pipeline
LAYER 1: ATTACK VECTORS
Unstructured Target Ingestion
• URLs & Deceptive Domains
• Phishing Emails & Urgency Lures
• Optical QR Codes (Quishing)
• Obfuscated Scripts & Shellcodes
• CISA / Threat Intel Advisories
• Enterprise Firewall Syslogs
• Client Password Inputs (Local)
LAYER 2: BOUNDARY WAF & ENGINE
Pre-Execution Interception & Filters
• Pattern-Based SQLi/XSS Filters
• Boundary WAF Neutralization Modal
• Shannon 8-Bit Entropy Calculation
• Levenshtein Typosquatting Match
• Cloudflare RFC 8484 DoH Query
• Cryptographic SPF/DMARC Audit
• Scikit-Learn TF-IDF + Naive Bayes
LAYER 3: FORENSIC & RESPONSE
Attribution & Mitigation Outputs
• Payload Dissector & Deobfuscator
• PowerShell UTF-16LE / Base64 Decode
• Automatic Defanging & IOC Extraction
• STIX 2.1 JSON & CSV Intelligence Export
• MITRE ATT&CK v14.1 Enterprise Map
• Gemini AI Incident Response Copilot
• Immutable SOC Logs & Telemetry DB
Built with Vanilla JS, Tailwind CSS, FastAPI Serverless, Scikit-Learn & Google Gemini AI | Author: Raviraj Parmar

1. Executive Overview & Problem Statement

Modern cyber adversaries utilize polymorphic obfuscation, DNS tunneling, multi-vector social engineering, and automated exploitation scripts to bypass legacy firewalls. Traditional academic cybersecurity projects often rely on superficial mockups, fake randomized counters, or basic forms that lack real analytical depth.

AI CyberShield was engineered to solve this dilemma by providing an authentic, commercial-grade Security Operations Center (SOC) platform. It merges client-side cryptographic telemetry with serverless heuristics and machine learning, equipping users with real tools utilized by Tier-1 and Tier-2 SOC analysts.

Core Platform Objectives:

- **Authentic Threat Detection:** Eliminate gimmicks and implement genuine heuristic formulas (Shannon 8-Bit Entropy, Levenshtein distance, RFC 8484 DNS queries).
- **Forensic Lab Capabilities:** Provide a live deobfuscation sandbox for Base64, UTF-16LE, and Hex shellcodes with automated defanging and MITRE technique attribution.
- **Threat Intelligence Ingestion:** Parse raw security bulletins, extract IOCs (IPs, Hashes, Domains, CVEs), and export directly to industry-standard STIX 2.1 JSON.
- **Standardized Taxonomy:** Map defensive capabilities directly to the MITRE ATT&CK;® Enterprise Matrix across all 7 primary kill-chain stages.
- **Zero-Exposure Privacy:** Execute cryptographic audits and password entropy calculations purely on the client side without transmitting sensitive cleartext.

2. System Architecture & Technical Specifications

AI CyberShield is built on a resilient, high-speed micro-architecture designed for instantaneous execution without heavy frontend framework overhead. The system operates across three interconnected layers:

Architecture Tier	Components & Technologies	Function in Pipeline
1. Presentation & Navigation	Vanilla JavaScript (ES6+), Tailwind CSS, Canvas API, FontAwesome 6	Renders collapsible sidebar, real-time Zulu clock, radar canvas, and reactive UI (<50ms response).
2. Analytical Heuristics & ML	Python FastAPI, Scikit-Learn TF-IDF, Shannon Entropy, Levenshtein Algorithm	Computes risk probabilities, detects brand typosquatting, evaluates linguistic urgency signals.
3. Network Forensics & CTI	Cloudflare RFC 8484 DoH, STIX 2.1 JSON Serializer, NIST NVD API	Resolves authoritative A/MX records, validates SPF/DMARC anti-spoofing, parses CVE identifiers.

4. Boundary WAF Defense	RegEx Injection Neutralizer, Client Token Vault (AES-256), SOC Audit Logger	Intercepts SQLi/XSS at boundary before query execution; alerts operator with forensic telemetry modal.

3. Deep Dive into Core Functional Modules

Each module within AI CyberShield serves a dedicated operational function within a modern Incident Response workflow. Below is the exact technical operation and sample workflow for each tool:

3.1 SOC Payload Dissector & Script Deobfuscator

Purpose: Adversaries heavily encode payloads to bypass static signature detection. This sandbox deconstructs multi-layer obfuscations into clean, actionable, defanged forensic plaintext.

Supported Encodings: Base64 (RFC 4648), PowerShell UTF-16LE, Hex Byte Streams (\x41\x42 / 0x...), and URL Percent-Encoding.

Forensic Mechanics:

- Automatically converts malicious protocols (http:// → hxxp://) and defangs domains (domain[.]com) to prevent accidental execution.
- Employs regex parsers to unearth embedded command-and-control (C2) IP addresses, stager URLs, and spawned system binaries (e.g., cmd.exe, powershell.exe, certutil).
- Maps the observed technique to the MITRE ATT&CK; taxonomy (e.g., **T1059.001 - PowerShell Stager** or **T1027 - Obfuscated Information**).

Example Scenario: Ingesting a Base64-encoded PowerShell stager immediately reveals an unauthorized WebClient DownloadString targeting an off-shore C2 server, tagging it as CRITICAL severity.

3.2 Threat Intel IOC Extractor & CVE Vulnerability Radar

Purpose: Security teams frequently receive unstructured incident reports, firewall dumps, or CISA advisories. This engine parses raw text and structures indicators of compromise (IOCs) into categorized threat intelligence.

Extraction Breakdown:

- **IP Addresses:** Differentiates between internal RFC1918 private IPs (10.0.0.0/8, 192.168.0.0/16) and external public WAN attack origins.
- **File Hashes:** Simultaneously detects MD5 (32 hex), SHA1 (40 hex), and SHA256 (64 hex) threat implant signatures.
- **Domains & FQDNs:** Extracts command nodes, strips sanitization brackets, and filters legitimate benign domains.
- **CVE Identifiers:** Identifies vulnerability CVE numbers and generates direct hyperlinks to the NIST National Vulnerability Database (NVD) for real-time CVSS scoring.
- **STIX 2.1 JSON & CSV Export:** Generates standard STIX 2.1 bundles with unique UUIDs (bundle--...) ready for SIEM ingestion (Splunk, Elastic, Sentinel).

3.3 Live Domain & Security Headers Forensic Inspector (RFC 8484)

Purpose: Executes live cryptographic domain lookups and perimeter posture evaluation without requiring server-side command execution.

Analytical Layers:

- **DoH DNS Resolution:** Queries Cloudflare DNS-over-HTTPS (RFC 8484) to resolve authoritative IPv4 (A) and Mail Exchanger (MX) records.

- **Cryptographic Anti-Spoofing:** Audits Sender Policy Framework (SPF) records and Domain-based Message Authentication (DMARC) policies to detect spoofability.
- **HTTP Security Headers Audit:** Evaluates HSTS, Content-Security-Policy (CSP), X-Frame-Options, and X-Content-Type-Options to issue a weighted grade from **A+ (Hardened)** to **F (Vulnerable)**.

3.4 MITRE ATT&CK;® Enterprise Matrix Navigator (v14.1)

Purpose: Connects real-world adversary behavior to standardized industry taxonomy. Provides interactive visibility across the 7 critical stages of enterprise compromise:

Tactic ID	Tactic Name	Key Enterprise Techniques & Defenses
TA0043	Reconnaissance	T1595 Active Scanning (Defended via WAF Rate Limiting & Header Obfuscation)
TA0001	Initial Access	T1566 Spearphishing, T1190 Exploit Public App (Defended via Heuristic Scanner)
TA0002	Execution	T1059 PowerShell & Command Shell (Defended via Payload Dissector Sandbox)
TA0003	Persistence	T1547 Registry Run Keys, T1053 Scheduled Tasks (Audited via SOC Telemetry)
TA0005	Defense Evasion	T1027 Obfuscated Files, T1562 Disable Security Tools (Deobfuscator Engine)
TA0006	Credential Access	T1003 OS Credential Dumping, T1110 Brute Force (Shannon Entropy Shield)
TA0011	Command & Control	T1071 Web Protocols, T1573 Encrypted C2 (Inspected via RFC 8484 DoH)

3.5 Password Sentinel (Shannon Cryptographic Entropy)

Mathematical Formulation: The entropy \(H\) of an input credential string \(S\) is computed using the Shannon 8-Bit Entropy equation:

$$H(X) = -\sum [P(x_i) \times \log_2(P(x_i))]$$

where \(P(x_i)\) represents the frequency probability of each unique character in the credential pool.

- Evaluates brute-force resistance across single-thread and GPU-cluster cracking scenarios.
- **Zero-Network Architecture:** Calculation executes 100% within the user's browser runtime. Cleartext credentials never cross network boundaries or reach remote logs.

3.6 Multi-Vector Threat Scanner (URL, Email, Optical QR)

- **URL Domain Heuristics:** Computes Shannon domain randomness and Levenshtein edit distance to identify lookalike brand domains (e.g., paypa1 vs paypal, micros0ft vs microsoft).
- **NLP Phishing Classifier:** Combines Scikit-Learn TF-IDF vectorization with heuristic urgency token matching (e.g., 'immediate suspension', 'wire transfer', 'OTP update') to predict phishing probabilities.
- **Optical Quishing Engine:** Ingests uploaded QR code image artifacts or camera feeds, extracts embedded URI strings, and funnels them directly into deep neural threat inspection.

3.7 Active Boundary WAF Interception Modal

Simulates an enterprise Web Application Firewall (WAF). When an operator submits attack signatures (such as SQL Injection ' OR 1=1-- or cross-site scripting <script>), the perimeter guard immediately neutralizes the payload at the client boundary, logs the incident with a forensic tracking token, and displays an alert modal detailing the blocked vector.

4. Oral Defense & Presentation Walkthrough Guide

When explaining or demonstrating AI CyberShield to examiners, teachers, or recruiters, follow this structured 5-minute presentation script for maximum impact:

Phase & Time	What to Say & Demonstrate (Talking Points)	Live Action on Screen
1. Introduction (0:00 - 1:00)	*'AI CyberShield is an enterprise-grade SOC platform designed to address real-world cyber threats like phishing, malicious payloads, and perimeter exploitation using authentic heuristics, DNS forensics, and MITRE ATT&CK; mapping.'*	Show live homepage, collapse/expand the left sidebar with Ctrl+B, point out the live UTC Military Zulu Clock.
2. Threat Scanner (1:00 - 2:00)	*'Notice our multi-vector scanner: we don't just check links; we analyze Shannon domain entropy, check lookalike brand spoofs, and process optical QR codes to stop quishing attacks before victims scan them.'*	Click 'Load Threat Sample' on Scanner, run inspection, show the SVG Circular Risk Meter animate to 88% High Risk.
3. Forensic Lab (2:00 - 3:30)	*'Here is our standout feature: the SOC Payload Dissector. When attackers encode PowerShell stagers in Base64 or Hex, analysts don't need external tools. CyberShield decodes it, extracts C2 IPs, defangs the URLs, and maps to MITRE T1059.'*	Click 'Payload Dissector', click 'PowerShell Stager', hit 'Execute Forensic Deobfuscation', show defanged output & extracted IOCs.
4. Threat Intel & MITRE (3:30 - 4:30)	*'Security analysts work with CISA bulletins. In our Threat Intel tab, we ingest unstructured advisories and generate STIX 2.1 JSON or CSV feeds in one click, fully aligned with the MITRE Enterprise Matrix.'*	Go to 'IOC Threat Intel', click 'Load Sample CISA Advisory', extract IOCs, click 'Export STIX 2.1'. Then switch to 'MITRE ATT&CK; Matrix'.
5. Conclusion (4:30 - 5:00)	*'CyberShield proves that modern web applications can combine client-side cryptographic speed with deep threat intelligence, providing zero-trust security without bloated dependencies.'*	Show live URL running on Vercel: https://cybershield-cyber-shield-raviraj-parmar.vercel.app/

5. Project Conclusion & Verification Links

AI CyberShield represents a complete, full-stack cybersecurity intelligence suite that meets enterprise standards. All components are active and accessible via the production deployment:

Production Deployment URL:	https://cybershield-cyber-shield-raviraj-parmar.vercel.app/
Source Code Repository:	https://github.com/ravirajsinhp011-star/cybershield
Deployment Platform:	Vercel Global Edge Network with CI/CD Auto-Deploy
Documentation Generation:	Automated ReportLab Enterprise Compiler (PDF + PPTX)